

Abdurrahman Wachid Shaffar

abdurrahman@shaffar.com | shaffar.com | linkedin.com/in/awachid | github.com/awshaff

TECHNICAL SKILLS

Programming: C++, Python, CUDA, CMake

Frameworks & Tools: Qiskit, QuTIP, Git, Docker

Research Areas: Quantum Microwave Communication, Continuous-Variable Quantum Information, Entanglement Distribution, Rydberg Atomic Receivers, Wireless Communication Systems, Quantum-Classical System Modeling

EXPERIENCE

Postdoctoral Associate

Apr. 2024 – Present

SPARK Lab, MIT (Advisor: Prof. Luca Carlone)

Cambridge, MA

- Reduced multi-session map inconsistency in city-scale environments by building a large-scale multi-robot SLAM system enabling real-time 3D mapping across 10+ robots and sessions.
- Accelerated research output by mentoring 4+ graduate researchers, contributing to 4+ published papers on 3D perception and autonomous mapping.
- Grew combined GitHub adoption to 4K+ stars by maintaining TEASER++ [🏆★2.1K], Hydra [🏆★773], KISS-Matcher [🏆★643], VGGT-SLAM [🏆★882].

Postdoctoral Fellow

Mar. 2023 – Mar. 2024

Korea Advanced Institute of Science and Technology (KAIST) (Advisor: Prof. Hyun Myung)

Daejeon, South Korea

- Improved localization robustness in dynamic scenes by developing real-time LiDAR-inertial and visual SLAM pipelines, deployed on 3+ mobile robot platforms.
- Reduced dynamic-object-induced map corruption by developing deep learning perception modules for moving object segmentation in crowded and unstructured environments.

Visiting Scholar

Nov. 2022 – Feb. 2023

StachnissLab, University of Bonn (Advisor: Prof. Cyrill Stachniss)

Bonn, Germany

- Cut dynamic-object artifacts in LiDAR maps by ~90% by building ERASOR2, an instance-aware removal system for static map building in urban scenes (RSS 2023).

Research Intern

Apr. 2022 – Sep. 2022

NAVER LABS

Gyeonggi-Do, Korea

- Achieved robust global LiDAR registration tolerating up to 99% outlier correspondences by developing Quatro, exploiting ground segmentation for reliable loop closure in degenerate urban environments (ICRA 2022 / IJRR 2023).

SELECTED PROJECTS

KISS-Matcher [🏆★643] *ICRA 2025* | Out-of-the-box fast and robust global point cloud registration for large-scale mapping.

VGGT-SLAM [🏆★882] *NeurIPS 2025* | Dense real-time RGB SLAM on SL(4) manifold; globally consistent 3D reconstruction.

Patchwork++ [🏆★842] *IROS 2022* | >100 Hz LiDAR ground segmentation; deployed in AV and mobile robot production.

ERASOR [🏆★494] *RA-L 2021* | ~94% dynamic-object artifact removal, enabling long-term autonomous navigation.

[e.g., **Real-Time LiDAR SLAM System**] | [Category, Year] – Reduced [X] by [Y]% in city-scale environments by [Z]; deployed on [N]+ platforms.

[e.g., **3D Object Detection Pipeline**] | [Category, Year] – Improved [X] recall by [Y]% by [Z], processing at [N] Hz on [hardware] (e.g., NVIDIA Orin).

PUBLICATIONS

10+ first-author papers at top-tier robotics and AI venues: IEEE ICRA, RSS, IEEE RA-L, NeurIPS, IEEE/CVF ICCV, IJRR

HONORS & AWARDS

2025 – Outstanding Reviewer Award, IEEE ICRA'25 (5 awardees out of 7,641 reviewers)

2024 – RSS Pioneers 2024 (30 selected out of 202 candidates)

2024 – 1st Prize, HILTI SLAM Challenge, IEEE ICRA'24

2023 – Best Paper Award, IEEE RA-L (5 of 1,100 papers)

2023 – 1st Prize, HILTI SLAM Challenge, IEEE ICRA'23 (63 international teams)

2023 – CES'23 Innovation Award (via technology transfer with HILLS Robotics)

2022 – 2nd Prize, HILTI SLAM Challenge, IEEE ICRA'22

2020 – Student Best Paper Award, ICCAS

EDUCATION

Kyung Hee University (KHU)

Yongin, South Korea

Ph.D. in Electronics and Information Convergence Engineering

Mar. 2022 – Feb. 2028

Universitas Gadjah Mada (UGM)

Yogyakarta, Indonesia

B.S. in Physics

Aug. 2015 – Nov. 2019